

Genetics and Heredity

DNA, genes, chromosomes, and inheritance patterns

This material is designed as a concise student handout. It introduces core ideas, important vocabulary, and short review questions for classroom or self-study use.

Learning outcomes

- Explain the core concepts using accurate biology terminology.
- Connect biological structures to their functions.
- Apply the concepts to simple examples and review questions.
- Use the vocabulary section to strengthen scientific communication.

1. DNA and genes

DNA is the molecule that stores genetic information. Sections of DNA that influence traits are called genes.

Genes contain instructions for making proteins or functional molecules. These instructions help determine inherited characteristics such as blood type, certain physical features, and many biological processes.

Key points

- DNA stores biological instructions.
- A gene is a segment of DNA associated with a trait or function.
- Proteins are built from instructions encoded in genes.

Mini activity

Write a short analogy that compares DNA to a recipe book and genes to individual recipes.

2. Chromosomes and inheritance

DNA is packaged into structures called chromosomes. In humans, most body cells contain 46 chromosomes arranged in 23 pairs.

During sexual reproduction, offspring inherit one chromosome from each pair from each parent. This is why offspring share traits with both parents but are not identical to either one.

Key points

- Chromosomes are organized DNA structures.
- Gametes carry half the usual chromosome number.
- Fertilization restores pairs of chromosomes.

Mini activity

Use colored pencils to draw two chromosome pairs and show how one chromosome in each pair can come from each parent.

3. Alleles and simple dominance

Different versions of the same gene are called alleles. Some alleles are dominant, meaning their effect can be seen when one copy is present.

A recessive allele usually affects the trait only when two copies are present. Punnett squares help predict possible allele combinations in offspring.

Key points

- Alleles are alternative forms of a gene.
- Dominant alleles can mask recessive alleles.
- Punnett squares show probability, not guaranteed outcomes.

Mini activity

Construct a Punnett square for Tt x Tt and calculate the possible genotype ratio.

Important vocabulary

Term	Meaning
DNA	Molecule that stores genetic instructions.
Gene	Segment of DNA linked to a biological function or trait.
Chromosome	Long DNA structure packaged with proteins.
Allele	Alternative version of a gene.
Genotype	Allele combination of an organism.
Phenotype	Observable trait or characteristic.

Quick review quiz

1. What is the relationship between DNA, genes, and chromosomes?
2. Why do offspring inherit traits from both parents?
3. What is the difference between genotype and phenotype?
4. In a simple dominant trait, what does recessive mean?
5. Why are Punnett square results probabilities rather than guarantees?

Teacher note: Students should answer the review questions before checking a textbook, notes, or class discussion.